PS 26165 – OIL SIF Precursor Detection
Professional Project Overview

Executive Summary

Build an AI-powered safety intelligence platform that reads Unsafe Act (UA), Unsafe Condition (UC), near-miss, and incident reports, identifies Serious Injury or Fatality (SIF) potential, maps reports to IOGP Life-Saving Rules, and highlights recurring safety risks through dashboards and analytics.

1. Problem Understanding

Oil India receives large numbers of safety observations and incident reports. Manual review is slow and important warning signs may be missed. The solution helps safety officers focus on high-risk events before a serious accident occurs.

Core Deliverables

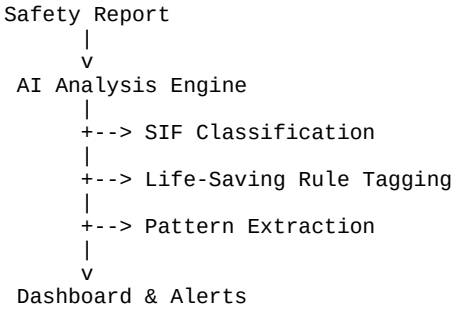
• Classify reports as High-SIF or Low-SIF potential.

• Tag reports with relevant IOGP Life-Saving Rules.

• Identify recurring risk patterns and hotspots.

• Provide an interactive dashboard for decision making.

System Flow



2. Recommended Tech Stack

Frontend: React.js, Tailwind CSS, Plotly/Recharts

Backend: FastAPI (Python)

Database: PostgreSQL

NLP: Sentence Transformers (MiniLM)

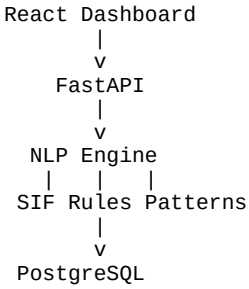
Classification: XGBoost / Logistic Regression

Rule Tagging: BART-MNLI Zero-Shot Classification

Entity Extraction: spaCy

Deployment: Docker

Architecture Diagram



3. 2-Day Prototype Roadmap

Day 1 Morning – Create 50–100 sample safety reports and define risk labels.

Day 1 Afternoon – Build SIF classification and rule-tagging logic.

Day 1 Evening – Develop a Streamlit or React demo interface.

Day 2 Morning – Build charts and analytics dashboard.

Day 2 Afternoon – Add explainability and polish the demo.

Demo Scenario

```
Input:
Worker stood under a suspended load.

Output:
Risk Level: HIGH
Rule: Safe Mechanical Lifting
Reason: Exposure to falling-object hazard
```

4. Winning Potential Analysis

Overall Rating: 8.5/10. The problem has strong real-world value, clear deliverables, and a compelling AI angle. Teams that understand SIF potential and safety-rule mapping can differentiate themselves significantly.

Strengths

• Real industrial safety impact

• Clear evaluation criteria

• Strong AI/NLP component

• Excellent dashboard demonstration potential

Risks

• Limited public datasets

• Requires understanding of safety terminology

• SIF potential is harder than normal severity classification

Professor Demo Pitch

Our system proactively detects Serious Injury and Fatality precursors from safety reports, maps them to Life-Saving Rules, and helps organizations prevent accidents before they occur.